CompoDigix

IIT Indore · STWUTEE Research Group

User Manual

Compost plant design and decision support

Recipe and mass balance · six composting systems · construction drawings · FCO quality grading · market price · Compost Mitra

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How to read this manual

The manual is organised in the order a new user meets the application. Chapter 1 explains what CompoDigix does and does not do. Chapter 2 covers access, languages and installation. Chapters 3–4 walk through planning a compost plant and reading every result tab, Chapter 5 covers the quality and market tool, and Chapter 6 the Compost Mitra helper. Chapter 8 is for the administrators, and Chapter 9 gives step-by-step hosting instructions written for readers with no server experience. Reference tables (FCO limits, feedstocks, glossary) are collected in the appendices.

Interface labels are printed in **this style**. Text in boxes marked **Note** gives practical advice; boxes marked **Caution** flag points that affect safety, legality or money.

Scope of the results

CompoDigix gives preliminary, indicative designs and estimates for planning. Structural design, statutory approvals and laboratory testing of the product remain the responsibility of the user and their engineer. The calculations run on the CompoDigix server; the manual describes what the results mean, not the internal equations.

CHAPTER 1

About CompoDigix

1.1 The problem it solves

Most farmers, institutions, municipal staff and small entrepreneurs who want to compost organic waste face the same questions: how much dry material to mix with wet waste, which composting system suits the site, how large the tanks, pits, piles or beds must be, what it will cost, and whether the product will be saleable. Answers are scattered across manuals, scheme guidelines and supplier brochures. CompoDigix brings them together in one guided workflow that starts from the waste a user actually has each day and ends with a buildable design and a priced product.

1.2 What CompoDigix does

1. **Balances the recipe.** From the daily quantity of each waste, the application adds the right amount of a dry carbon material, recycled finished compost, wood chips (as a re-usable bulking agent) and water, so that the fresh mix reaches a suitable carbon-to-nitrogen (C:N) ratio, moisture and free air space.
2. **Closes the mass balance.** Every kilogram entering the plant is accounted for leaving it: compost, water vapour, carbon dioxide, ammonia, leachate, recycled compost and screened wood chips. Inputs and outputs agree to within a fraction of a per cent.
3. **Compares six composting systems** — NADEP tank, Indore pit, windrow, aerated static pile, rotary drum and vermicompost beds — and scores each for the site (scale, land, power, climate, odour risk, labour, product and economics).
4. **Produces field-ready designs:** a dimensioned section and plan for the chosen system, a dimension schedule, a site layout drawn to scale, an equipment list and a batch schedule adjusted for the coldest month of the state.
5. **Grades the product** against the Fertiliser Control Order (FCO) 1985, Schedule IV, gives a quality index (0–100) and a realistic market price, and advises how to improve both. An optional AI reviewer can explain the result in the user's language.
6. **Prices the project:** capital cost, yearly operating cost and income, net present value, internal rate of return, payback and cost per tonne of waste.
7. **Helps in five languages** — English, Hindi, Bengali, Marathi and Gujarati — with voice read-out and the Compost Mitra chat helper.

1.3 Who it is for

User	Typical use
Farmers and households	NADEP tanks or pits for dung and crop residues; a farm-use product.
Gaushalas, dairies, poultry units	Windrows or vermicompost; selling surplus compost.
Institutions, campuses, housing societies	On-site processing of kitchen waste (bulk-generator duty under the SWM Rules 2016), usually a rotary drum or NADEP.
Urban local bodies	Town-scale windrow or aerated static pile plants; FCO city compost.
Entrepreneurs, FPOs, SHGs	Business plans for vermicompost or FOM; prices and payback.
Researchers and students	Mass balances, sensitivity of designs to feedstock and climate.

1.4 What CompoDigix does not do

- It does not replace a structural engineer: wall thicknesses, slabs and roofs shown in the drawings are standard practice, not a structural design.
- It does not replace a laboratory: pH, electrical conductivity and heavy metals cannot be predicted and must be measured before compost is sold as fertiliser.
- It does not issue approvals: authorisation from the State Pollution Control Board and fertiliser registration are obtained from the authorities.

CHAPTER 2

Getting started

2.1 Opening the application

CompoDigix runs in any modern web browser on a phone, tablet or computer. It is opened from the **CompoDigix** button (brown, beside the green BioDigix button) at the top of the STWUTEE Lab website, or directly at the `/compodigix/` address of the website.

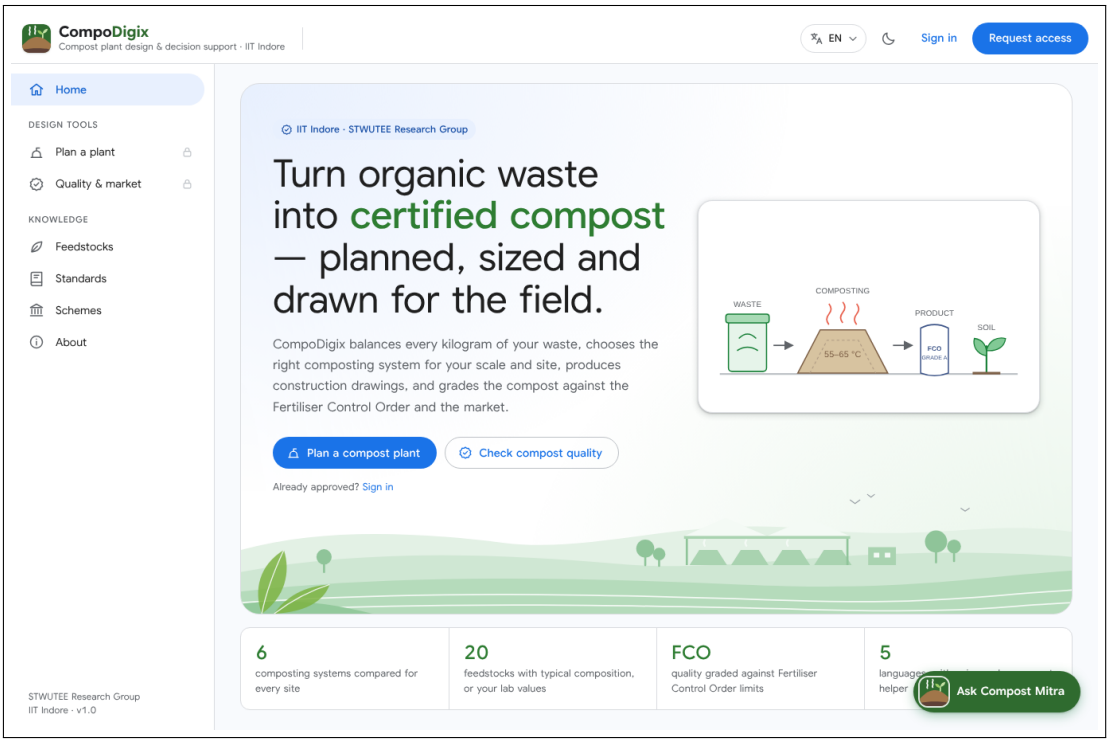


Figure 2.1. The CompoDigix welcome page. The two main actions are **Plan a compost plant** and **Check compost quality**.

2.2 Requesting access

The **Feedstocks**, **Standards**, **Schemes** and **About** pages are open to everyone. The two design tools need an approved account.

1. Select **Request access**.
2. Enter name, e-mail, a password of at least eight characters, the user type (farmer, institution, urban local body and so on), organisation, state and a short description of the intended use.
3. Submit. The request reaches the administrators at IIT Indore by e-mail.
4. After approval, sign in with the same e-mail and password. A signed-in session lasts 30 days on that device.

Note

While a request is pending, the dashboard says so and offers the open knowledge pages. Administrators' own accounts are approved automatically.

2.3 Language, theme and installation

- **Language.** The language button in the top bar switches all text, figures and reports between English, हिन्दी, বাংলা, मराठी and ગુજરાતી. The choice is remembered on the device.
- **Night view.** The moon button switches between light and dark themes; by default the phone or computer setting is followed.
- **Install as an app.** On phones, the browser menu offers **Add to Home screen** or **Install app**; on computers, the install icon in the address bar. The installed app opens without the browser frame and keeps working when the connection drops (calculations still need the internet).
- **Listen.** The speaker button on a result reads it aloud in the chosen language.



Figure 2.2. CompoDigix on a phone in night view, in Hindi.

CHAPTER 3

Planning a compost plant

3.1 Overview of the planner

The planner (**Plan a plant**) has an input column on the left and the result on the right. Results update automatically a moment after any input changes. On phones the result follows the inputs; the **See results** button jumps to it.

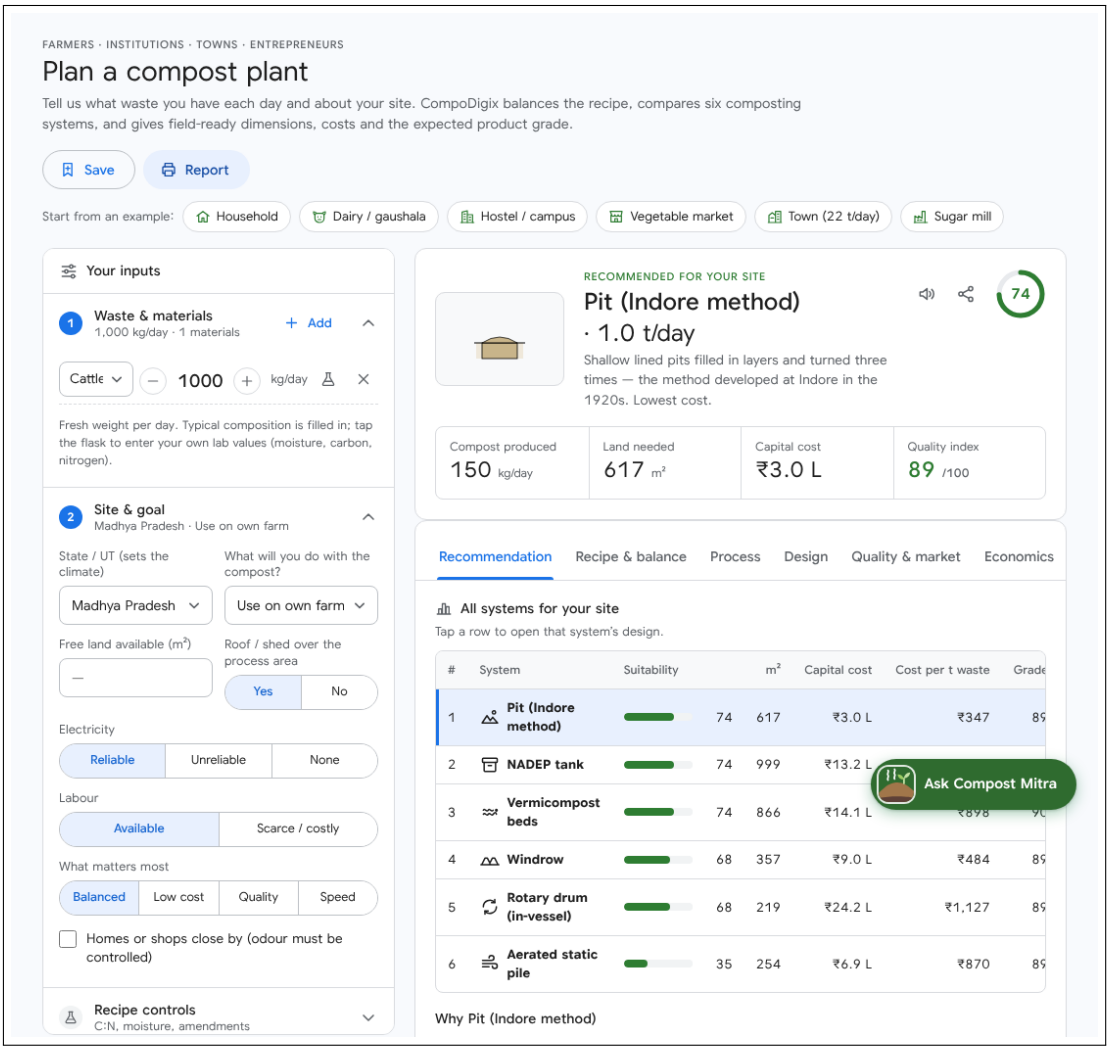


Figure 3.1. The planner. Inputs on the left are grouped in four numbered sections; the example buttons above them load ready-made cases.

3.2 Starting from an example

Six buttons under the title load typical cases that can then be edited:

Example	Waste and setting
Household	20 kg/day kitchen waste; homes close by; use on own garden.
Dairy / gaushala	1 t/day cattle dung; farm use.
Hostel / campus	400 kg/day food waste with 60 kg/day grass; homes close by; FCO city compost.
Vegetable market	3 t/day market waste; homes close by; FCO city compost.
Town (22 t/day)	18 t/day segregated municipal wet waste with 4 t/day market waste.
Sugar mill	20 t/day press mud with 5 t/day dewatered biogas slurry; FOM.

3.3 Section 1 — Waste and materials

Each row is one waste with its fresh weight per day. Use **Add** for more rows and the cross to remove one. Twenty common materials are listed (Appendix B). Typical moisture, carbon and nitrogen are filled in automatically. When laboratory values are available, select the flask icon on the row and enter **Moisture %**, **Carbon % dry** and **Nitrogen % dry**; these replace the typical values for that row.

Weighing the waste

Weigh the waste for at least a week and use the average day. Seasonal wastes (mango season, festival flowers, harvest residues) are best planned for the busiest month.

3.4 Section 2 — Site and goal

State / UT
sets the monthly air temperatures used to stretch the composting time in cold months and to check that earthworms will stay active.

What will you do with the compost?
Use on own farm, *Sell as city / organic compost (FCO)*, *Sell as vermicompost (FCO)* or *Sell as fermented organic manure (FOM)*. This sets the target product class, moisture at dispatch, testing costs and prices.

Free land available
in square metres, if limited. Systems that need more than the available land are marked down or ruled out.

Roof / shed
whether the process area is roofed (recommended in the monsoon; it adds cost).

Electricity
reliable, unreliable or none. Aerated piles and drums need power.

Labour
available or scarce; manual systems are marked down when labour is scarce.

What matters most
balanced, low cost, quality or speed.

Homes or shops close by
favours enclosed or aerated systems that control odour.

3.5 Section 3 — Recipe controls

By default (**Balance it for me**) the application chooses the additions. The controls allow an expert to steer the recipe:

- **Dry carbon material to add** — paddy straw, wheat straw, dry leaves, sugarcane trash, sawdust, coir pith or rice husk (or none).
- **Nitrogen-rich material to add** — used only when the waste is too woody.
- **Target C:N of fresh mix** (default 28) and **Target moisture** (default 55 %).
- **Mix back some finished compost** — recycled compost dries very wet waste without raising the C:N and seeds microbes.
- **Use wood chips for air space** — added only where the free air space would fall below 30 %; they are screened out and reused.
- **Use my mix as entered** — switches the balancing off; warnings then show what is wrong with the mix.

3.6 Section 4 — System, money and prices

- **Composting system** — *Recommend for me* (default) or a fixed choice.
- **Windrow turning** — automatic choice, by hand, or by a windrow turner.
- **Tipping fee received** and **Disposal cost avoided**, in rupees per tonne of waste.
- **Discount rate**, **Labour wage** and **Electricity** tariff.
- **Extend curing until C:N meets FCO** — curing is lengthened (up to 60 days) when the product C:N would otherwise exceed 20.
- **Sell in 50 kg bags**, **Eligible for Market Development Assistance**, and own selling prices per tonne. Left blank, the current rates kept by the administrators are used; the date of those rates is shown under the fields.

CHAPTER 4

Reading the results

The result card shows the recommended system with a suitability ring (0–100), the daily quantity, a short description and four key figures: **Compost produced**, **Land needed**, **Capital cost** and **Quality index**. Warnings about the mix appear under the figures. Seven tabs give the detail.

4.1 Recommendation

The table lists all six systems with suitability, land, capital cost, cost per tonne of waste and product grade. Selecting a row opens that system's full design; **Back to the recommended system** returns. The list under *Why …* gives the reasons behind the score in plain language (for example, *Needs electricity, which is not available*). Figure 1 in the tab compares suitability and levelised cost.

How the six systems compare

System	Comfortable range of fresh waste	Batch time (warm months)	Notes
NADEP tank	20 kg – 2 t/day	about 3 months	Brick tank, filled in layers and sealed; no turning; no power.
Pit (Indore method)	20 kg – 1.5 t/day	about 3 months	Lowest cost; three turnings; needs dry ground.
Windrow	0.5 – 300 t/day	2 months + curing	CPHEEO standard for towns; paved pad, leachate drain, turning.
Aerated static pile	2 – 200 t/day	3–4 weeks + curing	Blower and perforated pipes; no turning; good odour control.
Rotary drum	50 kg – 5 t/day	5 days + curing	Compact, enclosed; suits campuses and societies; needs power.
Vermicompost beds	50 kg – 20 t/day	2–3 weeks pre-composting + 2 months	High-value product; shade, moisture and labour essential.

When two systems score the same, the one with the lower cost per tonne is recommended.

4.2 Recipe and balance

The upper table is the balanced daily recipe: each material, the amount, moisture, C:N and bulk density, with chips marking what CompoDigix added (*added*, *recycled*, *structure*). Coloured chips summarise the fresh-mix C:N, moisture, free air space and bulk density against the targets (C:N 25–30 for fresh feed, moisture 50–60 %, free air space at least 30 %).

The mass-balance figure draws inputs and outputs per day to the same scale, and the two tables list them per day and per year. Recycled compost and wood chips appear on both sides because they circulate inside the plant. The panel below reports how closely the balance closes, the organic matter decomposed, nitrogen lost as ammonia, minimum air needed, heat released, compost yield and any extra drying needed on a yard before bagging.

RECOMMENDED FOR YOUR SITE

Pit (Indore method) · 1.0 t/day

74

Shallow lined pits filled in layers and turned three times — the method developed at Indore in the 1920s. Lowest cost.

Compost produced	Land needed	Capital cost	Quality index
150 kg/day	617 m²	₹3.0 L	89 /100

Recommendation | Recipe & balance | Process | Design | Quality & market | Economics

All systems for your site

Tap a row to open that system's design.

#	System	Suitability		m²	Capital cost	Cost per t waste	Grade
1	**Pit (Indore method)**		74	617	₹3.0 L	₹347	89
2	**NADEP tank**		74	999	₹13.2 L		
3	**Vermicompost beds**		74	866	₹14.1 L	₹898	9C
4	**Windrow**		68	357	₹9.0 L	₹484	89
5	**Rotary drum (in-vessel)**		68	219	₹24.2 L	₹1,127	89
6	**Aerated static pile**		35	254	₹6.9 L	₹870	89

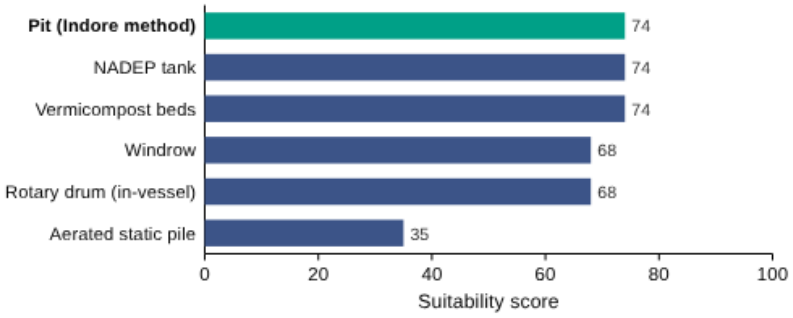
Ask Compost Mitra

Why Pit (Indore method)

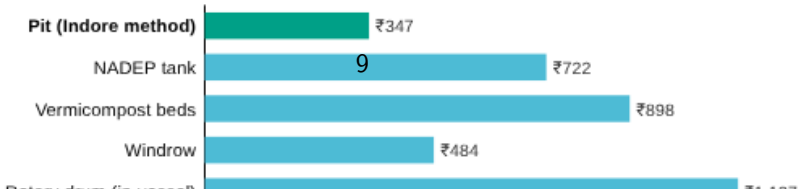
- Well suited to this amount of waste
- Simple to build and run with local skills

SVG PNG

a



b



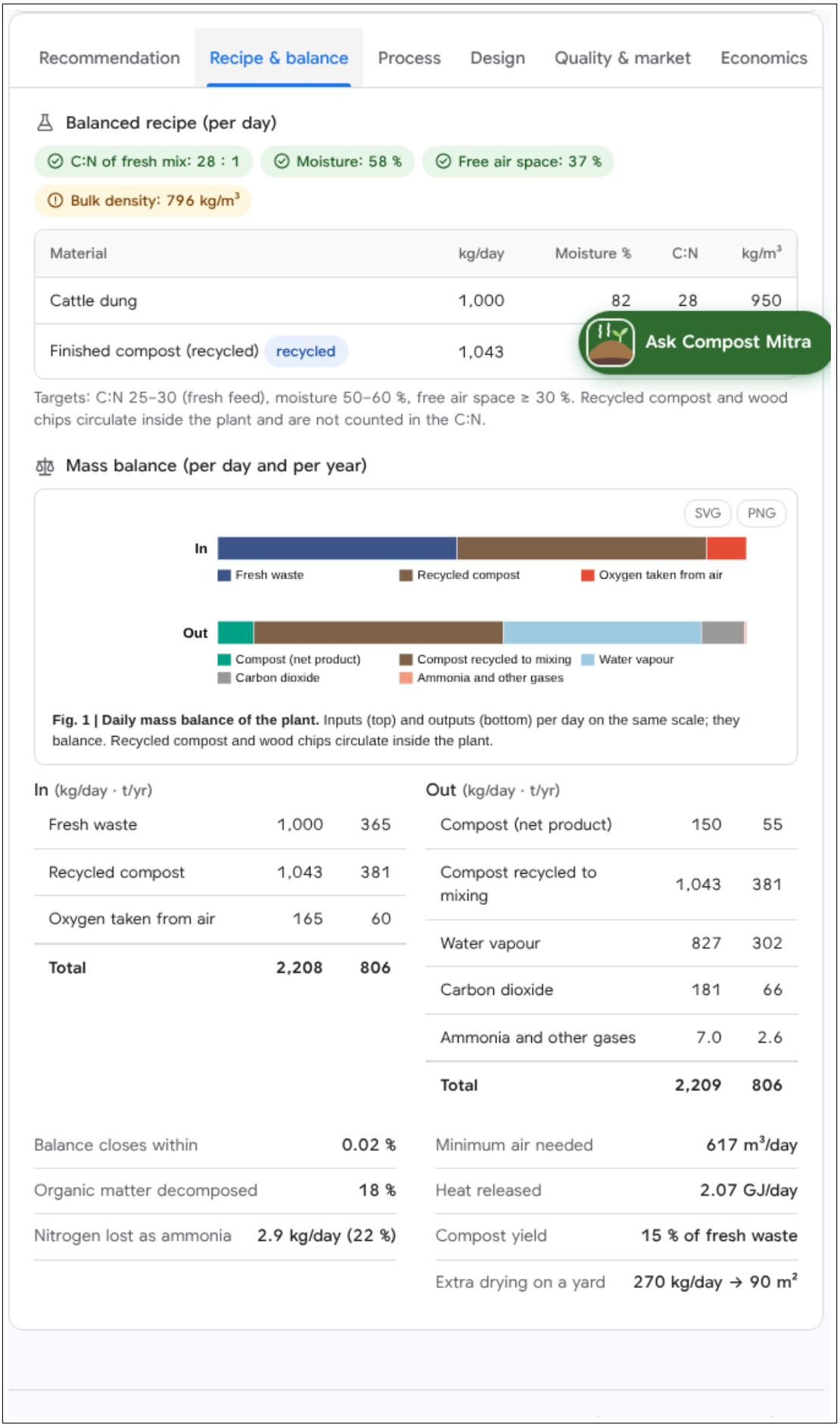


Figure 4.2. Recipe and balance tab.

4.3 Process

The timeline (Figure 1 of the tab) shows pre-composting, the active hot phase and curing for one batch; triangles mark turnings or key operations. Durations are stretched for the coldest month of the selected state, and Figure 2 shows that state's monthly air temperature, with months below 15 °C in red. The batch schedule lists what to do on which day, and *Targets to watch* lists the field checks (temperature, squeeze test, smell, shrinkage, maturity).

4.4 Design

- **Construction drawing** — panel *a* is a section and panel *b* a plan, drawn to scale with scale bars. Lettered dimensions (A, B, C …) are listed in the **Dimension schedule**. For many identical units only the first few are drawn; the count is in the schedule.
- **Construction note** — materials and key details for that system (for example, PCC M15 pad with 1–2 % slope for windrows; graded orifices facing down for aerated piles).
- **Area** — process units, receiving and sorting, product store (30 days), drying yard, office and laboratory, and the total including roads and drains.
- **Equipment** — weighbridge or platform scale, shredder, trommel screen, loader, windrow turner, blowers, controller or drums, sized to the fresh waste per day.
- **Workers per day**, **electricity** and **leachate tank** volume.
- **Site layout** (for plants needing 150 m² or more) — blocks drawn to area in flow order; small blocks are labelled with call-outs.

Both drawings can be downloaded as SVG (for CAD or editing) or PNG with the buttons on each figure.

Caution

The drawings follow standard practice and published norms (CPHEEO 2016, NABARD vermicompost models, BioCycle aerated-pile practice). A qualified engineer should check foundations, slabs, roofs and drainage for the actual site before construction.

4.5 Quality and market

This tab grades the predicted product of the selected system against the FCO class chosen in Section 2 (vermicompost for the vermicompost system). Composition values marked with an asterisk are predicted from the mass balance; pH, conductivity and heavy metals are shown as *Test needed*. The grade, the estimated price and the advice are explained in Chapter 5.

4.6 Economics

The four headline figures are the net present value (red when negative), internal rate of return, payback period and cost per tonne of waste treated. Below them are itemised capital cost, operating cost and income, and the cumulative cash position by year (capital in year 0; half of the equipment replaced in year 8).

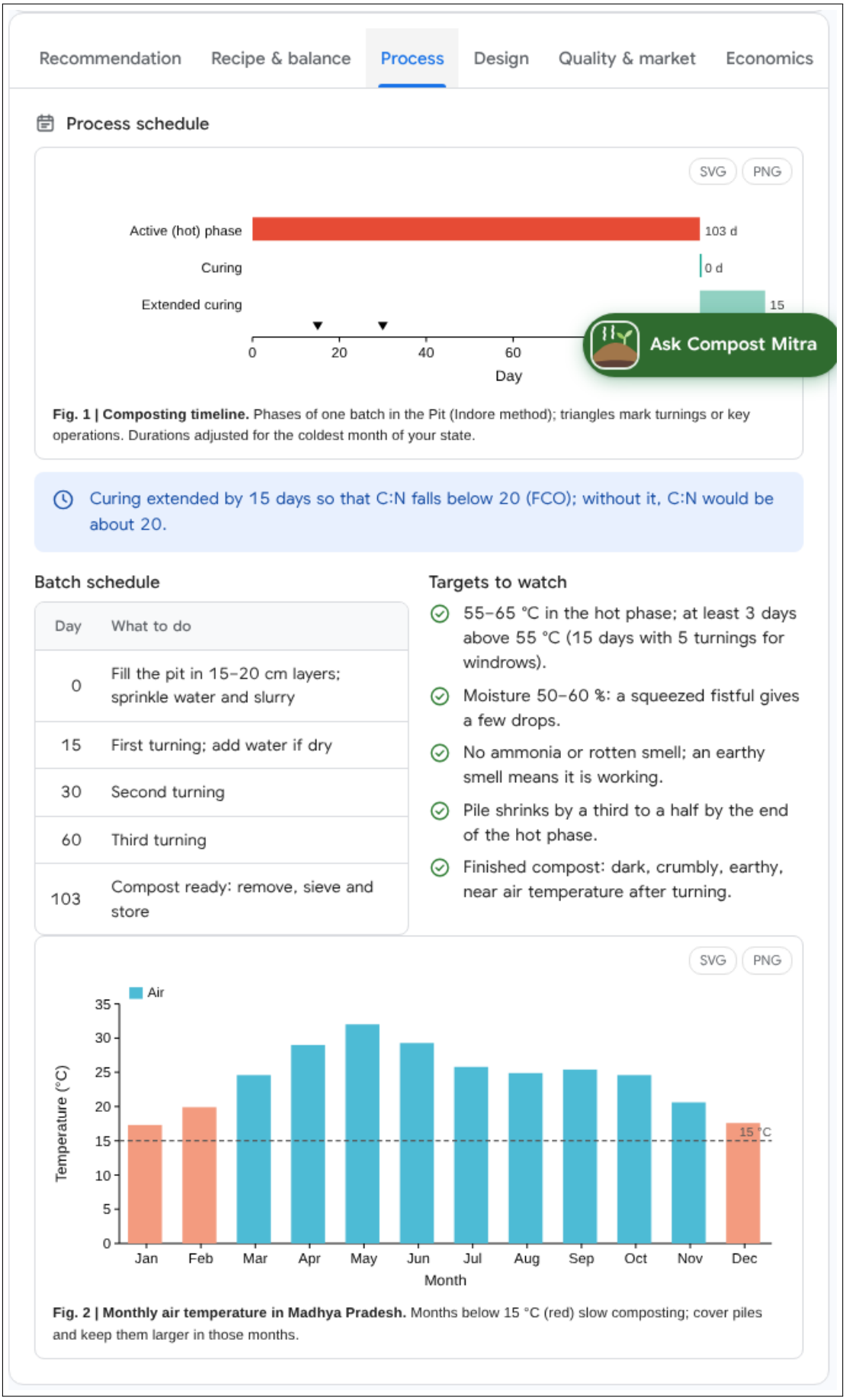
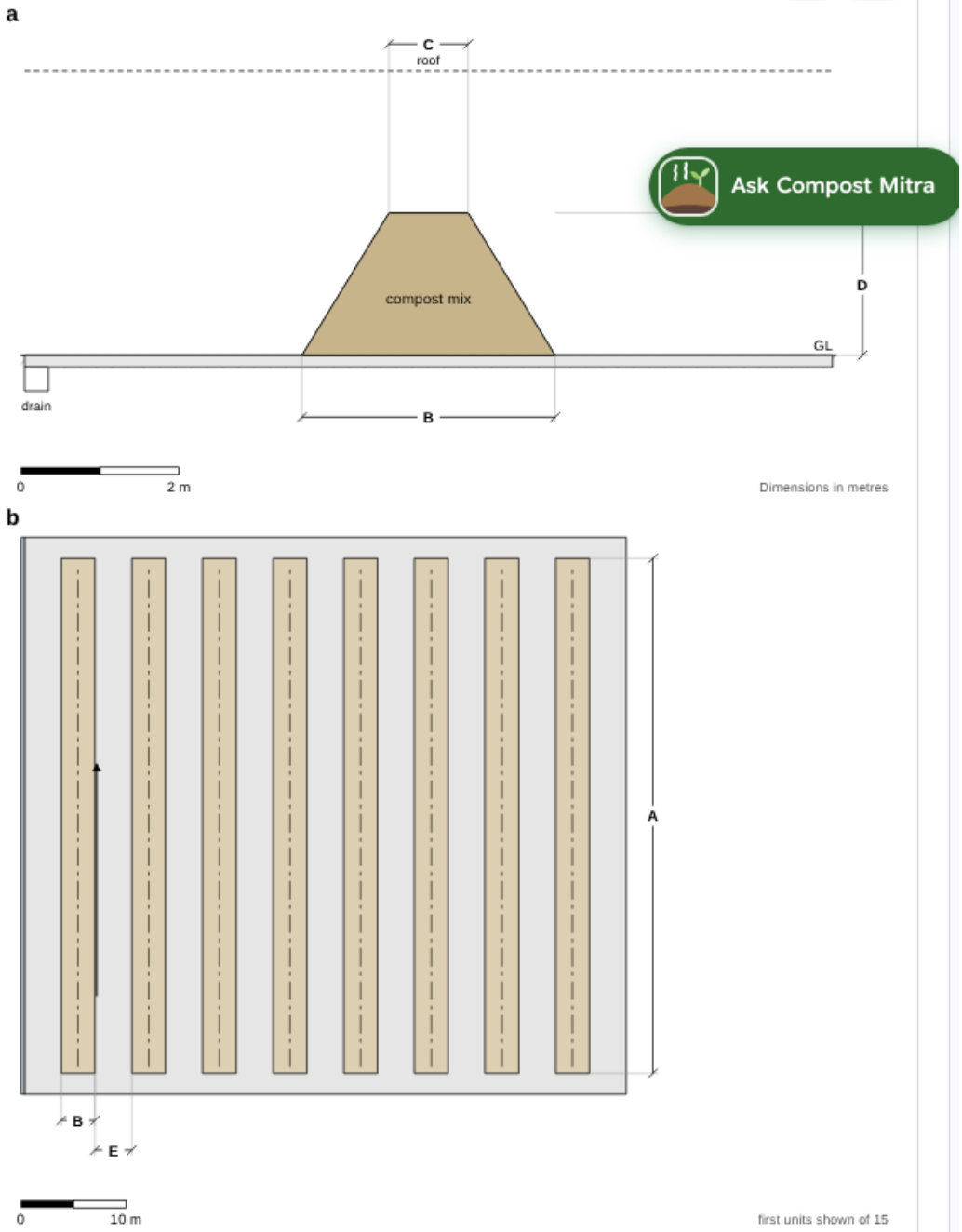


Figure 4.3. Process tab: timeline, batch schedule, field targets and the state's monthly temperatures.

Recommendation | Recipe & balance | Process | Design | Quality & market | Economics

Construction drawing

SVG PNG



Ask Compost Mitra

Fig. 1 | Construction drawing of the Windrow. a, Section. b, Plan (first units shown). Drawn to scale (see scale bars); dimensions in metres; lettered dimensions are listed in the schedule.

Dimension schedule (m)

Mark	Item	Value
A	Windrow length	48.9
B	Base width	3.2
C	Top width	1
D	Height	1.8
E	Aisle	3.5
	Number of windrows (active)	15

Reading a negative NPV

Small farm units often show a negative NPV because the compost is valued only at its fertiliser worth while labour is paid in cash. Where the family does the work, or where a tipping fee or avoided disposal cost applies, enter those figures in Section 4 — the result can change a great deal. Unit rates are indicative 2025–26 values and should be replaced with local quotations.

4.7 Build and operate

Construction notes for the chosen system, a daily operation checklist, a troubleshooting table (smell, cold pile, overheating, flies, leachate, dry pile), health-and-safety points, and the permits and standards that apply.

4.8 Saving, reports and sharing

- **Save** stores the inputs under a name on the server; saved plans are listed on the dashboard.
- **Report** prints or saves a PDF with an IIT Indore header, a unique report number (CDX-P-… for plans, CDX-Q-… for quality checks), a QR code that reopens the same plan, all tabs and three signature lines.
- The share button on the result sends a plain-language summary and link by WhatsApp or copies the link. Anyone with an approved account who opens the link sees the same plan.

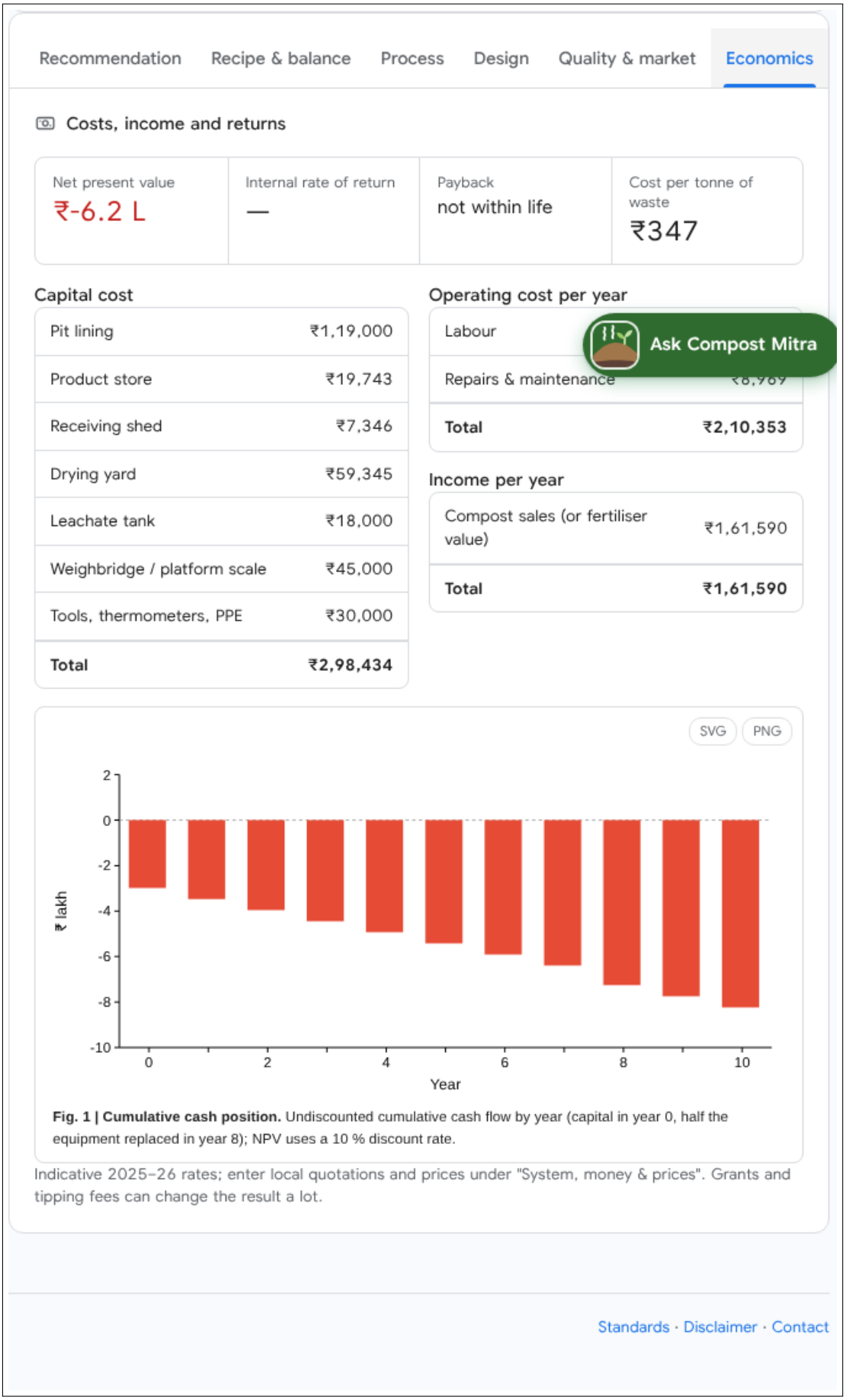


Figure 4.5. Economics tab.

CHAPTER 5

Compost quality and market

5.1 Entering a laboratory report

Open **Quality & market**, choose the product class (city compost, vermicompost, FOM or PROM) and type the values from the laboratory report: moisture, total organic carbon, nitrogen, phosphate (P_2O_5), potash (K_2O), pH, conductivity, bulk density and, under **Heavy metals**, arsenic, cadmium, chromium, copper, mercury, nickel, lead and zinc. Composition is on a dry-weight basis. C:N and N+P+K are calculated when their parts are entered. Tick **Pile stayed above 55 °C for at least 3 days** if temperature records show pathogen kill.

5.2 The grade

Each parameter is marked *Meets*, *Borderline* (predicted values only, within 10 % of the limit), *Fails* or *Test needed*. The quality index (0–100) combines FCO compliance, nutrient content, organic carbon and hygiene (pathogen kill and maturity). Grades: **A** 85 and above, **B** 70–84, **C** 55–69, **D** below 55. The lower table shows how the same sample would fare in each of the four FCO classes.

5.3 The price

A sample that meets its FCO class is priced from the current bulk rate for that class, adjusted up or down with the quality index; branded 50 kg bags earn more. A sample that fails any FCO parameter cannot legally be sold as fertiliser and is priced as a soil conditioner. The administrators keep the base rates up to date in the backend (Section 8.3).

5.4 Advice and the AI reviewer

The built-in advisor lists what is within limits, what is not, the market route (bulk price, bag price, likely buyers), specific improvements (curing longer, drying, segregation, enrichment, screening) and the next steps (notified laboratory, registration, Market Development Assistance for GOBARdhan FOM). When the administrators have switched AI on, **Ask AI advisor** sends the graded data — not the user's identity — to the AI service and returns a short review in the selected language. If the AI does not answer, the built-in advice stays on screen.

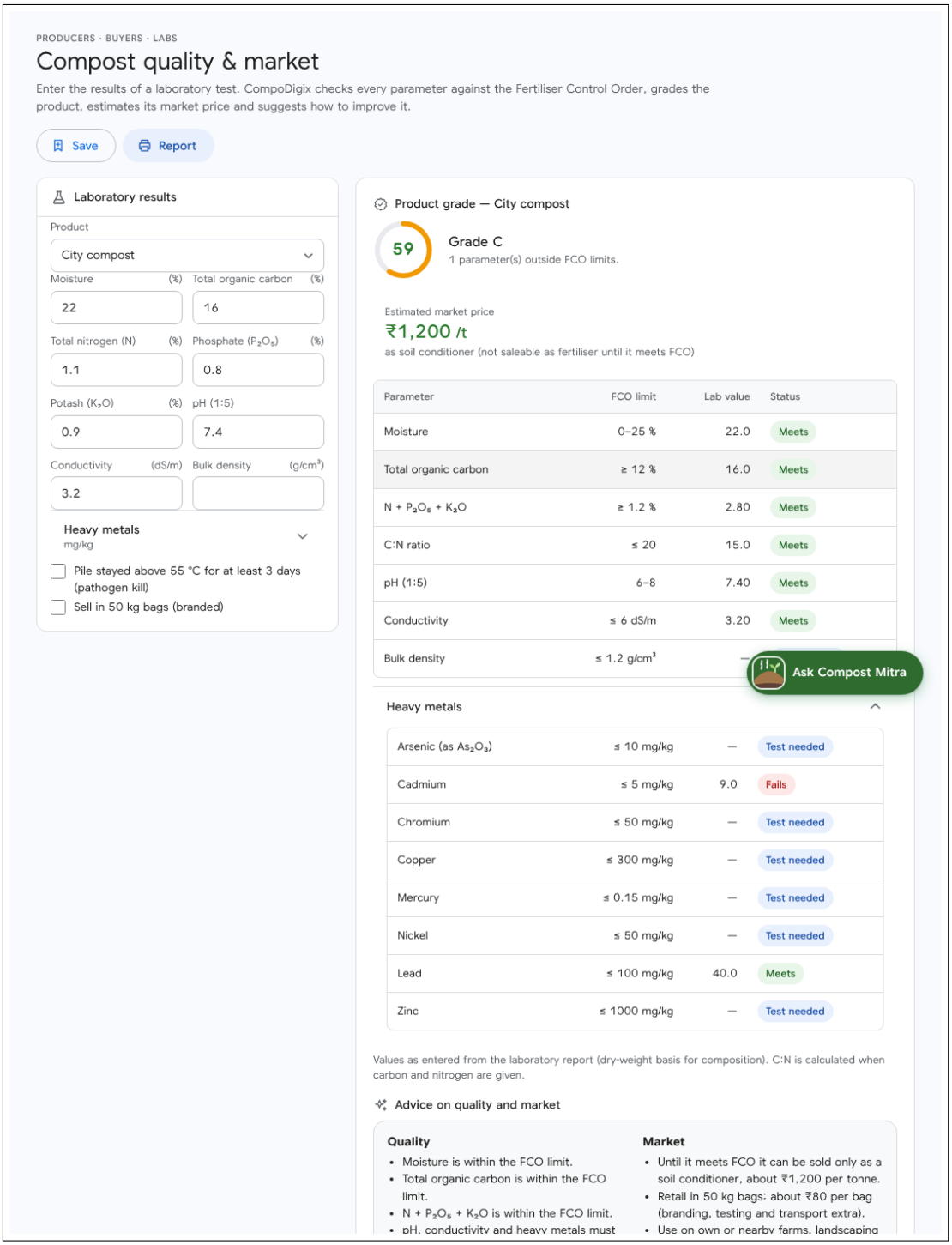


Figure 5.1. Quality and market tool with a sample that fails the cadmium limit.

CHAPTER 6

Compost Mitra

Compost Mitra is the chat helper opened from the **Ask Compost Mitra** button. It answers questions on making compost, the C:N ratio, moisture, temperature, smell, flies, slow or wet piles, turning, vermicompost, maturity, quality, prices, subsidies, costs, field application, safety, municipal rules and using the app. Questions can be typed or spoken in any of the five languages; answers can be read aloud. Buttons under an answer open the relevant part of the app — for example, a question such as *"500 kg a day — how much land?"* opens the planner with 500 kg/day already filled in. *"Explain my plan"* summarises the current plan.

Without AI, Compost Mitra answers from its built-in knowledge base. With AI switched on, it answers more freely and uses the current plan as context.

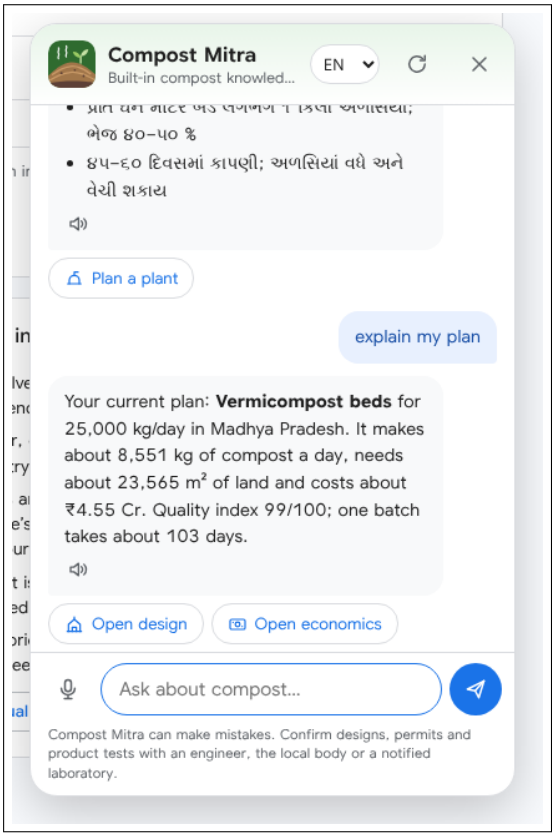


Figure 6.1. Compost Mitra answering in Gujarati and explaining the current plan.

Caution

Compost Mitra can make mistakes. Designs, permits and product tests should be confirmed with an engineer, the local body or a notified laboratory.

CHAPTER 7

Knowledge pages

Feedstocks
typical moisture, carbon, nitrogen and C:N of twenty materials, a C:N figure, and rules of thumb for greens and browns.

Standards
FCO Schedule IV summary for city compost, vermicompost, FOM and PROM; heavy-metal limits; process standards (pathogen kill, cycle length, moisture, pad and drainage) and the rules that apply (SWM Rules 2016, FCO 1985, CPHEEO manual).

Schemes
Market Development Assistance, GOBARdhan, Swachh Bharat Mission (Urban) 2.0, the SWM Rules 2016 duties of bulk generators, the city-compost policy and PKVY.

About
inventors, methods in brief, citation (text and BibTeX), disclaimer and the user manual link.

KNOWLEDGE BASE

Feedstock library

Typical composition of common wastes. Mix nitrogen-rich "greens" with carbon-rich "browns" to reach a C:N of about 25–30. Measure your own materials for a real project.

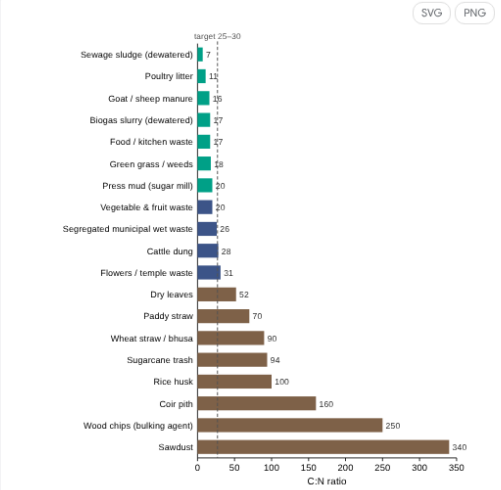


Fig. 1 | Carbon-to-nitrogen ratio of common feedstocks. Green bars are nitrogen-rich (C:N below 20), blue are near the target, brown are carbon-rich. The dashed line marks the target for a compost mix.

Rules of thumb

- Greens: food waste, vegetable waste, dung, grass — wet and rich in nitrogen.
- Browns: straw, dry leaves, sawdust, husk — dry and rich in carbon.
- Squeeze test: a fistful should feel like a wrung-out sponge — a few drops, no stream.
- Chop to 2–5 cm. Keep out plastic, glass, metal, batteries and medical waste.

Ask Compost Mitra

Material	Moisture %	C %	N %	C:N	Role	Note
Food / kitchen waste	75	48	2.8	17	Green (N-rich)	Very wet, nitrogen-rich, breaks down fast; mix with dry browns at once to avoid smell.
Vegetable & fruit waste	85	47	2.3	20	Brown (C-rich)	Wettest common waste; needs a lot of dry material or recycled compost.
Segregated municipal wet waste	60	34	1.3	26	Brown (C-rich)	Only source-segregated wet waste; plastics, glass and metals lower quality and add heavy metals.
Flowers / temple waste	60	46	1.5	31	Brown (C-rich)	Remove plastic, threads and foil; marigold stalks are slow — shred them.
Cattle dung	82	44	1.6	28	Brown (C-rich)	Ideal starter: near-balanced, full of microbes; add straw for structure.
Poultry litter	30	38	3.5	11	Green (N-rich)	Strong nitrogen source; loses ammonia if not mixed with carbon-rich material.
Goat / sheep manure	50	41	2.5	16	Green (N-rich)	Dry pellets, rich in nutrients; soak or mix with wet waste.
Green grass / weeds	75	46	2.6	18	Green (N-	Green and wet: wilt for a day so it does not mat down.

Figure 7.1. Feedstock library.

CHAPTER 8

For administrators

Administrators are the accounts listed in the backend (mayur.jain@iiti.ac.in and stwuteeres@iiti.ac.in). They see an extra **Admin** page.

8.1 Approving users

New requests arrive by e-mail and appear under **Pending**. Approve or decline each one; the person is told by e-mail. Approved users can be moved back to declined at any time. **CSV** downloads the user list.

8.2 Usage insights

The admin page shows registered and approved users, plans and quality checks run, waste planned, a three-panel figure (users by state; plans and checks per week; systems recommended) and Compost Mitra statistics, including the latest questions that were not understood — useful for improving the knowledge base.

8.3 Updating prices and unit rates

All prices and unit rates live in the **Settings** tab of the Google Sheet *CompoDigix data*:

- `price.city`, `price.vermi`, `price.fom`, `price.prom`, `price.soil` — rupees per tonne, bulk, at the plant gate.
- `unit.wage`, `unit.power`, `unit.diesel`, `unit.pad`, `unit.roof`, `unit.turnerT` and the other `unit.` rows — local rates in rupees.
- `prices.updated` — the date shown to users beside the rates.
- `mitra.ai` — `on` or `off` for AI answers.

Edit only the value column. Changes reach users within ten minutes.

8.4 Switching on AI (optional)

1. Obtain an API key from Anthropic (Claude) or Google (Gemini).
2. Open the backend project at `script.google.com` → **Project Settings** (gear icon) → **Script properties** → **Add script property**.
3. Property `MITRA_API_KEY`, value: the key. Save.

The provider is recognised from the key. AI use is billed by the provider to the key's owner. Removing the property, or setting `mitra.ai` to `off`, switches AI off; the built-in answers continue to work.

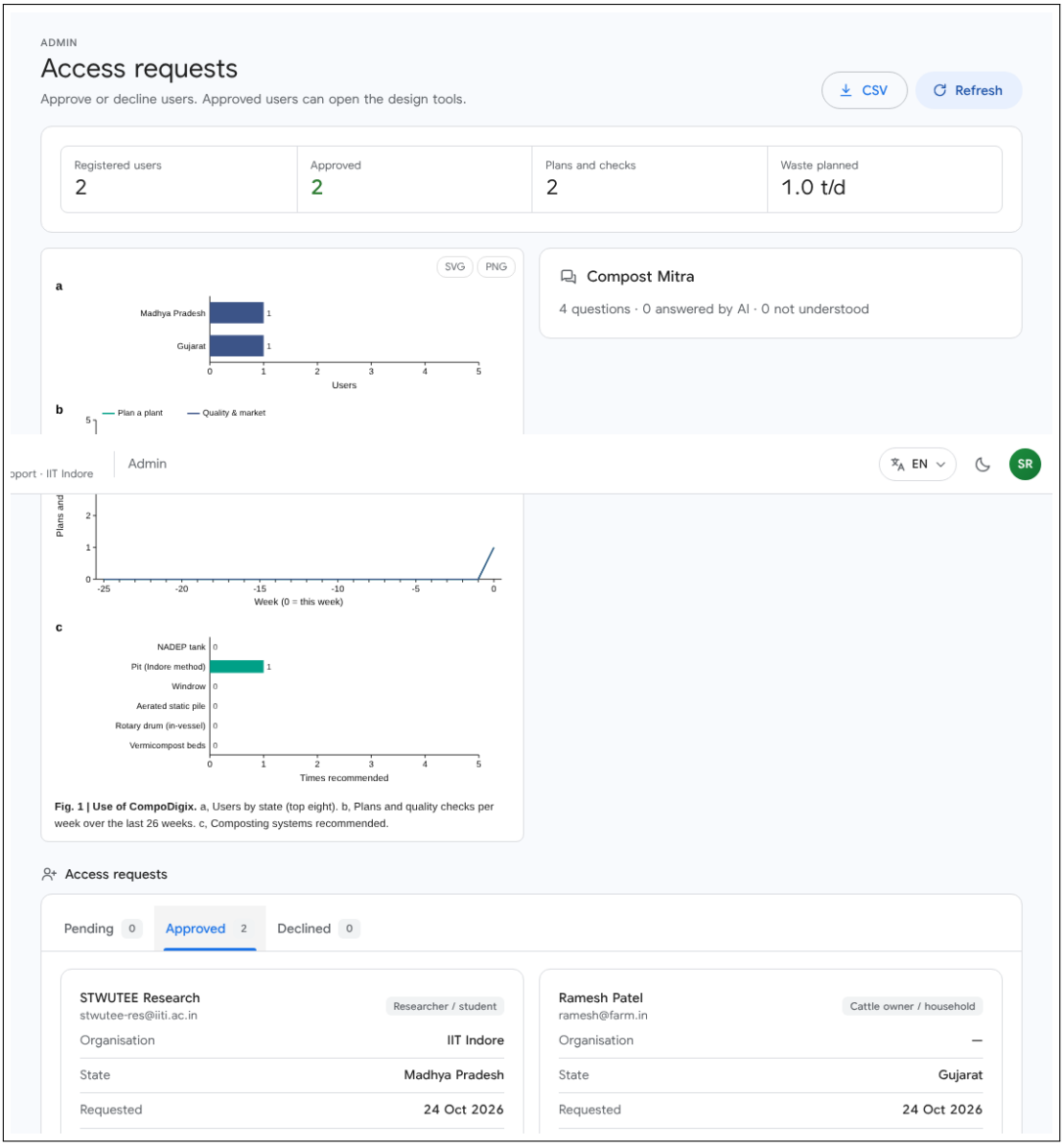


Figure 8.1. Admin page with usage insights and access requests.

CHAPTER 9

Hosting and installation

CompoDigix has two parts: the **website files** (folder `compodigix`) on the lab web server, and the **backend** — a Google Apps Script web app with its own Google Sheet — which does the calculations and keeps accounts. The backend is completely separate from the STWUTEE lab portal and from BioDigix.

9.1 Step A — create the backend (once, about 10 minutes)

1. Sign in to Google as `mayur.jain@iiti.ac.in` or `stwutee-res@iiti.ac.in`.
2. Open `script.google.com` → **New project**. Rename it *CompoDigix backend*.
3. Delete the sample code. Open the file `CompoDigix.gs`, copy everything, paste it in, and select **Save**.
4. In the function list at the top choose `setup` → **Run**. Allow the permissions Google asks for. A sheet called *CompoDigix data* appears in Google Drive.
5. **Deploy** → **New deployment** → gear icon → **Web app**. Set **Execute as: Me** and **Who has access: Anyone**. Select **Deploy** and copy the **Web app URL** (it ends in `/exec`).

9.2 Step B — put the website files online

1. Open the file `compodigix/config.json` in a text editor and paste the URL between the quotes:
   `{ "backendUrl": "https://script.google.com/macros/s/.../exec" }`
2. In CloudPanel, open the site → **File Manager** → the site's `htdocs` folder.
3. Upload the `compodigix` folder next to the existing `biodigix` folder (or upload the full lab-website package, which already contains it, together with the updated `index.html` that carries the new CompoDigix button).
4. Open `https://`*`lab-website`*`/compodigix/` and request access with an administrator e-mail; it is approved automatically.

9.3 Updating later

- **New website files:** upload the new `compodigix` folder over the old one, keeping `config.json`.
- **New backend file:** paste the new `CompoDigix.gs`, run `setup` once, then **Deploy** → **Manage deployments** → pencil icon → **Version: New version** → **Deploy**. The URL stays the same.

9.4 Troubleshooting the application

What is seen	What to do
Yellow banner *Preview mode*	`config.json` has no backend address. Complete Step B.1.
Could not reach the CompoDigix server	Paste the `/exec` address into a private (incognito) browser window. It must show `{"ok":true,"service":"CompoDigix backend"}`. A Google sign-in page or *access denied* means the deployment is not set to **Who has access: Anyone** — change it under **Deploy** → **Manage deployments**. An address ending in `/dev`, or the editor address, is the wrong one: copy the **Web app URL** ending in `/exec`.
The CompoDigix backend is not set up yet	Run `setup` in the Apps Script project (Step A.4).
Sign-in works but results do not appear	Check the internet connection; select **Try again**. If it persists, redeploy the backend as a *new version*.
No e-mail on new requests	The backend sends mail from the account that deployed it; check that account's Sent folder and spam.
Prices look out of date	Edit the **Settings** tab (Section 8.3).
Ask AI advisor says AI is not available	Add `MITRA_API_KEY` and check `mitra.ai` is `on`.
Old version still showing after an update	Close and reopen the app, or refresh twice; the app updates itself in the background.

CHAPTER A

FCO specifications used for grading

Summary of the Fertiliser Control Order 1985, Schedule IV, as used by CompoDigix (dry-weight basis for composition). The latest notification should always be checked before sale.

Parameter	City compost	Vermicompost	FOM	PROM
Moisture (%)	≤25	15–25	≤40	≤25
Total organic carbon (%)	≥12	≥18	≥14	≥8
Total nitrogen (%)	—	≥1.0	—	≥0.4
Phosphate, P_2O_5 (%)	—	≥0.8	—	≥8.0
Potash, K_2O (%)	—	≥0.8	—	—
N + P_2O_5 + K_2O (%)	≥1.2	—	—	—
C:N ratio	<20	<20	<20	<20
pH (1:5)	6.0–8.0	6.5–7.5	6.0–8.4	6.5–8.0
Conductivity (dS/m)	≤6	≤4	≤4	≤10
Bulk density (g/cm³)	<1.2	0.65–0.95	—	<1.6
Particle size	90 % < 4 mm	90 % < 4 mm	—	90 % < 4 mm

Heavy-metal limits (mg/kg, all classes): arsenic (as As_2O_3) 10, cadmium 5, chromium 50, copper 300, mercury 0.15, nickel 50, lead 100, zinc 1000.

CHAPTER B

Feedstock library

Typical values (C and N on dry matter) used when no laboratory values are entered.

Material	Moisture %	C %	N %	C:N	Role
Food / kitchen waste	75	48	2.8	17	green
Vegetable and fruit waste	85	47	2.3	20	green
Segregated municipal wet waste	60	34	1.3	26	green
Flowers / temple waste	60	46	1.5	31	brown
Cattle dung	82	44	1.6	28	green
Poultry litter	30	38	3.5	11	green
Goat / sheep manure	50	41	2.5	16	green
Green grass / weeds	75	46	2.6	18	green
Press mud (sugar mill)	70	39	2.0	20	green
Biogas slurry (dewatered)	75	34	2.0	17	green
Sewage sludge (dewatered)	80	32	4.5	7	green
Dry leaves	30	47	0.9	52	brown
Paddy straw	12	42	0.6	70	brown
Wheat straw / bhusa	10	45	0.5	90	brown
Sugarcane trash	30	47	0.5	94	brown
Sawdust	25	51	0.15	340	brown
Rice husk	10	40	0.4	100	brown
Coir pith	15	48	0.3	160	brown
Wood chips (bulking agent)	35	50	0.2	250	structure

Sources: CPHEEO Manual on Municipal Solid Waste Management (2016); On-Farm Composting Handbook, NRAES-54 (1992); Indian field studies. Real projects should test their own materials.

CHAPTER C

Field troubleshooting

Sign	Likely cause	What to do
Rotten-egg or sour smell	Too wet or packed; no air	Turn now; add dry straw or recycled compost; check drainage.
Pile does not heat up	Too dry, too small, or too little nitrogen	Add water while turning; make the pile bigger; add dung or food waste.
Above 70 °C or white ash-like layer	Pile too big or too dry; fire risk	Turn and add water; reduce the pile height.
Flies, maggots or rats	Fresh food waste exposed	Bury fresh waste inside; cover with 15–30 cm compost or straw; keep the site clean.
Liquid running out	Too wet, or rain on an open pile	Add dry material; cover or roof; collect leachate and spray it on dry piles.
Dry, dusty, nothing happening	Moisture below 40 %	Water while turning until the squeeze test gives a few drops; shade the pile.
Ammonia smell	C:N too low	Add straw, leaves or sawdust.

CHAPTER D

Glossary

C:N ratio
Carbon-to-nitrogen ratio, by weight on dry matter. Fresh mixes work best at about 25–30; finished compost should be below 20.

Free air space (FAS)
Share of the pile volume filled with air; at least 30 % keeps the pile aerobic.

Bulking agent
Coarse, dry material (wood chips) that keeps the pile open; screened out and reused.

Curing
The cooler stage after the hot phase in which compost stabilises and matures.

Leachate
Liquid that drains from wet waste or rain on piles; it must be collected, not released.

Thermophilic (hot) phase
Stage at 55–65 °C in which pathogens and weed seeds are destroyed.

FCO
Fertiliser Control Order 1985; Schedule IV sets the specifications for organic fertilisers.

FOM / PROM
Fermented organic manure / phosphate-rich organic manure — FCO organic fertiliser classes.

NPV, IRR, payback
Net present value, internal rate of return and the years until cumulative cash turns positive.

Cost per tonne
Annualised capital plus yearly operating cost divided by tonnes of waste treated per year.

CHAPTER E

Citation and acknowledgements

Jain, M. S. & Chakravorty, S. (2026). *CompoDigix: compost plant design and decision-support system* (version 1.0) [Web application]. STWUTEE Research Group, Indian Institute of Technology Indore.

Design norms draw on the CPHEEO Manual on Municipal Solid Waste Management (2016), the Fertiliser Control Order 1985 and its amendments, NABARD vermicompost unit models, the On-Farm Composting Handbook (NRAES-54) and BioCycle guidance on aerated static piles. Climate data are monthly normals for each state.

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